Michele Fumagalli

Professor
University of Milano Bicocca
Department of Physics
Piazza della Scienza 3, 20126 Milano (MI), Italy
Email: michele.fumagalli@unimib.it

Scientific Interests

Gas flows around galaxies, galaxy formation and evolution, the role of environment, absorption line systems, physics of the interstellar medium, star formation, stellar initial mass function.

Academic History

2020- **Professor**, *University of Milano Bicocca*.
2020- **Associate**, *INAF - Osservatorio Astronomico di Trieste*.
2020-2023 **Visiting Professor**, *Durham University*.
2018-2020 **Professor**, *Durham University*.
2017-2018 **Associate Professor (Reader)**, *Durham University*.
2014-2017 **Assistant Professor (Lecturer)**, *Durham University*.
2013-2014 **Postdoctoral Fellow**, *Carnegie Observatories, Princeton University*.

Education

2016 **Postgraduate Certificate in Academic Practice**, *Durham University, UK*.
2012 **Ph.D. in astrophysics**, *University of California, Santa Cruz, USA*.
2010 **Master in astrophysics**, *University of California, Santa Cruz, USA*.
2008 **Laurea specialistica (MSc)**, *University of Milano Bicocca, Italy*.
2006 **Laurea triennale (BSc)**, *University of Milano Bicocca, Italy*.

Selected Awards and Fellowships

2017 **Abilitazione Nazionale Italiana**, *Professore Associato e Ordinario*.
2015 **Fellow of the Higher Education Academy**.
2014-2015 **Carnegie Visiting Associate**, Carnegie Observatories.
Visiting fellowship at Carnegie Observatories.
2012 **Lyman Spitzer Fellowship**, Princeton University.
Postdoctoral fellowship in theoretical astrophysics.
2012 **Carnegie-Princeton Fellowship**, Carnegie Observatories, Princeton University.
Postdoctoral fellowship in observational astrophysics.

2012 **Hubble Fellowship**, Carnegie Observatories.
Awarded to highly qualified recent postdoctoral scientists to conduct independent research.

2012 **CfA Fellowship**, (declined), The Harvard-Smithsonian Center for Astrophysics.
Awarded to an outstanding researcher displaying significant promise in theory or observation.

2012 **Miller Research Fellowship**, (declined), University of California, Berkeley.
Awarded to exceptional young scientists of great promise.

2011 **Price Prize in Cosmology and AstroParticle Physics**, CCAPP, Ohio State University.
Awarded in recognition of research excellence in cosmology and astro-particle physics.

2011 **Chancellor's Dissertation Year Fellowship**, UCSC.
Awarded based on the academic achievement of the nominee.

2010 **Whitford Prize**, Department of Astronomy, UCSC.
Awarded for outstanding performance during the first and second years.

2008 **Regents' fellowship**, UCSC.
Awarded to promising first-year graduate students.

Grant History (principal investigator or primary co-investigator)

2023 **Bando Prin 2022 PNRR**, MUR, (Node PI).

2022 **Dipartimenti di Eccellenza 2023-2027**, MUR, (Deputy PI).

2020 **Durham Astronomy Consolidated Grant**, STFC, (Project co-PI).

2019 **NASA grant**, HST-GO-15637, (Science Co-PI).

2018 **ERC Attrattività**, Fondazione Cariplo, (PI).

2017 **ERC Starting Grant**, ERC, (PI).

2017 **Durham Astronomy Consolidated Grant**, STFC, (Project PI).

2015 **NASA grant**, HST-GO-14127, (Science PI).

2012 **NASA Hubble Fellowship**, grant HF-51305.01-A, (PI).

2010 **HIPACC grant**, University California, (PI).

Talks, Seminars, Conferences

Jul., 2026 **The Many Scales of Galaxy Environments**, *Ascona, Switzerland*, Invited review.
From halos to the cosmic web: probing the gas ecosystem at z>3.

Jan., 2026 **UNIMI Colloquium Series**, *Milano, Italy*, Invited.
The gas environment of galaxies across 10 billion years.

Oct., 2025 **Multi-phase, Multi-temperature and Complex**, *Olbia, Italy*, Invited review.
The multiphase nature of the CGM: warm ionized gas in absorption and emission.

Aug., 2025 **Ascona meeting on cosmic ISM and Stellar Populations**, *Ascona, Switzerland*, Invited.
Mapping the cosmic baryon cycle over 10 billion years in the MUSE era

Nov., 2024 **A decade of discoveries with MUSE and beyond**, *ESO*, Invited Review.
The MUSE contribution to mapping the cosmic baryon cycle over 10 billion years

Sep., 2024 **ILR workshop at Osaka**, *Osaka, Japan*, Invited.
The gas environment of galaxies across 10 billion years

Jun., 2024 **IFPU Focus Week: Where the CircumGalactic medium meets the galaxy environment**, *Italy*, SOC member.

May, 2024 **OPINAS Colloquium**, *Max-Planck for Extraterrestrial Physics*, invited.
The gas environment of galaxies across 10 billion years

Apr., 2024 **Extreme Galaxies in their Extreme Environments at Extremely Early Epochs**, *Iceland*, SOC member.

Mar., 2024 **Bologna Joint Astrophysical Colloquium**, *INAF/University of Bologna, IT*, invited.
The gas environment of galaxies across 10 billion years

Jan., 2024 **Astronomy colloquium**, *Lancaster, UK*, invited.
The gas environment of galaxies across 10 billion years

Sep., 2023 **IMAGING 2023**, *Italy*, Organizing committee.
Connecting the dots between the CGM and the larger-scale environment

May., 2023 **IASF Colloquium Series**, *IASF-INAF, Milan*, Invited.
The gas environment of galaxies across 10 billion years

Apr., 2023 **The Department of Astronomy Colloquium Series**, *Tsinghua University*, Invited.
The gas environment of galaxies across 10 billion years

Mar., 2023 **The Circum-Galactic Medium across cosmic time: an observational and modeling challenge**, *The 52nd "Saas-Fee Advanced Course"*, Invited lecturer.
The multiphase CGM in absorption and emission

Feb., 2023 **The Multiphase Circumgalactic Medium**, *Ringberg Castle, Germany*, Invited.
The gas environment of galaxies across 10 billion years

Nov., 2022 **IoA colloquium**, *Cambridge, UK*, invited.
The gas environment of galaxies across 10 billion years

Sep., 2022 **What Matter(s) Around Galaxies 2022**, *Italy*, SOC chair and lead organizer.
Connecting the dots between the CGM and the larger-scale environment

Jun., 2021 **KIAA Forum on Gas in Galaxies for Early Career Scientists**, *Kavli Institute for Astronomy and Astrophysics*, invited.
Flows around galaxies: advancements, challenges and opportunities

May., 2021 **Multi-object Spectroscopy for Statistical Measures of Galaxy Evolution Workshop**, *Space Telescope Science Institute*, invited review.
Studying gas flows around galaxies with multi object spectroscopy

May., 2021 **Extragalactic Seminar Series**, *University of Victoria*, invited.
Shedding light on gas around galaxies across cosmic times

Apr., 2021 **Astronomy Colloquium**, *University of California, Santa Cruz*, invited.
Shedding light on gas around galaxies across cosmic times

Mar., 2021 **Physics Colloquium**, *North Carolina State University*, invited.
Shedding light on gas around galaxies across cosmic times

Nov., 2020 **Kapteyn Institute Colloquium**, *University of Groningen*, invited.
Shedding light on gas around galaxies across cosmic times

Jun., 2020 **EAS 2020**, *Leiden*, invited review.
Flows around galaxies in 2020: advancements, challenges and opportunities

Jun., 2020 **Insights into the CGM and ICM**, *IAP, France*, invited.
MUSE observations of the CGM of distant galaxies

Apr., 2020 **Astronomy Colloquium**, *Royal Observatory/Edinburgh*, invited.
Shedding light on gas around galaxies across cosmic times

Mar., 2020 **Astronomy Colloquium**, *INAF/Arcetri*, invited.
Shedding light on gas around galaxies across cosmic times

Dec., 2019 **Joint Astronomy Colloquium**, *MPA-MPE-ESO*, invited.
Shedding light on gas around galaxies across cosmic times

Oct., 2019 **CGM in Berlin 2019**, *Max Planck Society*, invited.
Gas around galaxies at $z \sim 2-3$: linking emission and absorption with large surveys

Jun., 2019 **What Matter(s) Between Galaxies**, *Abbazia di Spineto*, SOC.
Gas around galaxies: connecting emission and absorption with large surveys

Mar., 2019 **Astronomy Seminar**, *Nottingham University*, invited.
Shedding light on gas around galaxies across cosmic time

Dec., 2018 **Twenty years of science at Bicocca**, *Milano-Bicocca University*, invited review.
Astrophysics ± 20: Deeper, Sharper, and Bigger

Nov., 2018 **CASTOR UV space observatory**, *The Royal Observatory Edinburgh*, invited review.
The galaxy-IGM connection

Jun., 2017 **What Matter(s) Around Galaxies**, *Durham University*, SOC/LOC co-chair.
Probing the gaseous environment of star-forming galaxies in absorption and emission

Apr., 2017 **Seminar, Department of Physics**, *University of Milano-Bicocca*, invited.
MUS(E)ing over gas flows as drivers of galaxy evolution

May., 2016 **Cavendish Astrophysics Seminar**, *University of Cambridge*, invited.
Gas flows as fuel for star formation: a spotlight on strong absorption line systems

Apr., 2016 **Astronomy Seminar**, *ETH Zurich*, invited.
Gas flows as fuel for star formation: a spotlight on strong absorption line systems

Mar., 2016 **Astronomy Seminar**, *Stockholm University*, invited.
Gas flows as fuel for star formation: a spotlight on strong absorption line systems

Sep., 2015 **Astronomy Seminar**, *INAF/Trieste*, invited.
Gas flows as fuel for star formation: a spotlight on strong absorption line systems

Jun., 2015 **IGM@50**, *INAF/Firenze*, invited.
Probing gas flows near galaxies: a spotlight on Lyman Limit Systems

Jun., 2014 **Intergalactic Matters**, *MPIA, Heidelberg*, invited.
A shot in the dark: the star formation rates of DLAs at $z \sim 2-3$

Apr., 2014 **Colorful galaxies: a conference for Peppo Gavazzi's birthday**, *Como, Italy*, invited.
Can we use Hα to trace star formation rates?

Apr., 2014 **Exploiting VST ATLAS... and its sister surveys**, *Durham University*, invited.
ATLAS search for Lyman Limit Systems in quasar pairs.

Mar., 2014 **Astronomy Friday Lunch Talks**, *Durham University*.
The importance of stochastic effects in stellar population synthesis.

Jan., 2014 **DEX meeting**, *Durham University*.
Investigations on the gaseous environment of distant galaxies.

Dec., 2013 **TAPIR seminar**, *Caltech*, invited.
Investigations on the gaseous environment of distant galaxies.

Oct., 2013 **Metal Production and Distribution in a Hierarchical Universe**, *Rencontres de l'Observatoire de Paris 2013 - ESO Workshop*, invited review.
IGM abundances in the high-redshift universe.

Aug., 2013 **Santa Cruz Galaxy Workshop**, *UCSC*.
Lyman limit systems and the circumgalactic medium at $z \sim 2-3$.

Jun., 2013 **Intergalactic Interactions**, *Higgs Centre, Edinburgh*, invited.
Lyman limit systems and the circumgalactic medium at $z \sim 2-3$.

Jun., 2013 **ENIGMA workshop**, *MPIA*, invited.
Lyman limit systems and the circumgalactic medium at $z \sim 2-3$.

Apr., 2013 **Lunch Talk**, *Carnegie Observatories*.
Beyond the disk: The role of halo gas in galaxy formation.

Mar., 2013 **Hubble Fellows Symposium**, *STScI, Baltimore*.
Optically-thick hydrogen in the z=3 universe

Dec., 2012 **University of Milano-Bicocca**, *Milan*, invited.
The gaseous environment of distant galaxies

Nov., 2012 **UT Astronomy Colloquium**, *Austin*, invited.
The gaseous environment of distant galaxies

Sep., 2012 **Keck Science Meeting**, *San Diego*.
Pristine gas two billion years after the Big Bang

Jun., 2012 **Metals in Tuscany**, *INAF/Firenze*, invited.
Pristine gas two billion years after the Big Bang

May., 2012 **Price Prize lecture**, *CCAPP Ohio State University*, invited.
Cosmology with absorption line systems

Apr., 2012 **Astronomy Colloquium**, *Osservatorio Astronomico di Brera*, invited.
Cosmology with absorption line systems

Mar., 2012 **Turbulence in Cosmic Structure Formation**, *Arizona State University*.
Detection of pristine gas two billion years after the Big Bang

Jan., 2012 **DARK Cake Meeting**, *DARK Cosmology Centre*.
Detecting cold accretion and metal poor gas around galaxies

Jan., 2012 **219th AAS Meeting**, *Austin, TX*.
Exploring the gas cycle in high-redshift galaxies: a joint effort of theory and observations

Dec., 2011 **Theory meeting of the Galaxy and Cosmology group**, *MPIA Heidelberg*.
Probing inflow in high-redshift galaxies

Oct., 2011 **Theoretical Astrophysics Center seminar**, *UC Berkeley*, invited.
Exploring the gas cycle in high-redshift galaxies: a joint effort of theory and observations

Oct., 2011 **Lunch Talk**, *Carnegie Observatories*.
Exploring the gas cycle in high-redshift galaxies: a joint effort of theory and observations

Oct., 2011 **Astronomy Tea Talk**, *Caltech*.
Exploring the gas cycle in high-redshift galaxies: a joint effort of theory and observations

Aug., 2011 **Santa Cruz galaxy workshop**, *Santa Cruz*.
Cold streams and primordial gas at high redshift

Jul., 2011 **Celebrating the career of A. Wolfe**, *Schloss Ringberg*, invited.
Detecting cold streams with absorption line systems

Jul., 2011 **MPIA**, *Heidelberg*.
Stochastic star formation and IMF (non) variation

Jun., 2011 **Odyssey of cosmic baryons**, *Marseille*.
Detecting cold streams with absorption line systems

Jun., 2011 **Gas in galaxies**, *Kloster Seeon, Germany*.
Detecting cold streams with absorption line systems

Dec., 2010 **CASS, UCSD**, *San Diego*.
Gas in and around galaxies

Aug., 2010 **Santa Cruz galaxy workshop**, *Santa Cruz*.
Gas in simulations of $z > 2$ galaxies

May, 2010 **Como+Milano+Heidelberg+Marseille**.
Images and simulations to connect gas and stars in $z > 2$ galaxies

Apr., 2010 **UCSC**, *Santa Cruz*.
Hunting gas and stars in galaxies across the Universe

Aug., 2009 **Santa Cruz Galaxy Workshop**, *Santa Cruz*.
A shot in the dark: probing galaxies giving rise to DLAs at $z > 2$

Aug., 2009 **UCSC Friday Lunch Talk**, *Santa Cruz*.
Molecular gas deficiency in HI poor galaxies

Jun., 2009 **University of Chicago**, *Chicago*.
A shot in the dark: imaging of DLAs

Mar., 2009 **Università dell'Insubria**, *Como, Italy*.
Star formation $z = 0 - 3$

Dec., 2008 **CASS, UCSD**, *San Diego*.
The star formation rate and gas content in local spiral galaxies

Jul., 2008 **Università di Milano-Bicocca**, *Milano, Italy*.
The relationship between gas content and star formation rate in spiral galaxies

Proposal History (principal investigator or primary co-investigator)

2026 JWST; 54 hours, cycle 5.
2024 ALMA; 32 hours, cycle 11.
2023 HST; 100 SNAP targets, cycle 31.
2022 ALMA; 9 hours, cycle 9.
2022 ESO/VLT; 16 hours, P109.
2021 ALMA; 9 hours, cycle 8.
2019 ESO/VLT; 25 hours, P105.
2019 Hubble Space Telescope; 8 orbits, cycle 27.
2019 JCMT/SCUBA-2; 30 hours, 2019B.
2018 Hubble Space Telescope; 90 orbits, cycle 26 (LP).
2017 ESO/VLT; 250 hours, P101 (LP).
2018 JCMT/SCUBA-2; 16 hours, 2018A.
2017 ESO/VLT; 36 hours, P100.
2017 JCMT/SCUBA-2; 9 hours, 2017B.
2016 ESO/VLT; 18 hours, P99.
2016 Keck Telescope; 2 nights, 2016B.
2016 Hubble Space Telescope; 96 orbits, cycle 24 (LP).
2016 JCMT/SCUBA-2; 9 hours, 2016B.
2016 Keck Telescope; 1 night, 2016A.
2016 WHT; 12 nights, 2016A.
2016 ESO/VLT; 106 hours, P97-100 (LP).
2015 WHT; 9 nights, 2015B.
2015 ESO/VLT; 9 hours, P96.
2015 Hubble Space Telescope; 55 orbits, cycle 23.
2014 ESO/VLT; 28 hours, P95.
2014 ESO/VLT; 5 hours, P94.
2014 Gemini-S Telescope; 30 hours, 2014A.
2014 Magellan Telescope; 4 nights, 2014A.
2013 Magellan Telescope; 5 nights, 2013B.
2013 Keck Telescope; 1 night, 2013B.
2012 Keck Telescope; 1 night, 2013A.
2012 Magellan Telescope; 4 nights, 2013A.

2012 Magellan Telescope; 4 nights, 2012B.

2011 IRAM 30m Telescope; 64 hours, 2011B.

Teaching and Advising

2024- Introduction to Galaxies; MSc at University of Milano-Bicocca.

2024- Didactics of Mathematics; School of Education, University of Milano-Bicocca.

2021-2026 Medical Physics; School of Medicine, University of Milano-Bicocca.

2019- Astrophysics Laboratory; MSc at University of Milano-Bicocca.

2018-2019 Radiative processes in astrophysics; PhD lecture series at Durham University.

2018 The role of baryonic process in galaxy formation and evolution; PhD lecture series at University of Milano-Bicocca.

2016-2017 PHYS2651: Physics in Society, BSc at Durham University.

2014-2019 PHYS1081: Introduction to Astronomy, BSc at Durham University.

2014-2018 PHYS1101: Discovery Skills in Physics, BSc at Durham University.

2009 Ay2: Overview of the Universe, BSc at UCSC.

PhD Students
- Mr. Michele Fulghieri (2025-), University of Milano-Bicocca.
- Mr. Davide Tornotti (2024-), University of Milano-Bicocca.
- Mr. Georg Herzog (2020-2024), University of Milano-Bicocca (PhD, 2024).
- Mr. Calvin Sykes (2017-2021), Durham University (PhD, 2021).
- Ms. Louise Welsh (2017-2021), Durham University (PhD, 2021).
- Mr. Ruari Mackenzie (2014-2018), Durham University (PhD, 2018).
- Mr. Greg Ashworth (2014-2018), Durham University (PhD, 2018).

PDRAs
- Dr. Francesco Pistis (2023-2026), University of Milano-Bicocca.
- Dr. Trystyn Berg (2021-2023), University of Milano-Bicocca.
- Dr. Louise Welsh (2021-2023), University of Milano-Bicocca.
- Dr. Alessia Longobardi (2021-2023), University of Milano-Bicocca.
- Dr. Emma Lofthouse (2018-2023), Durham University, University of Milano-Bicocca.
- Dr. Rajeshwari Dutta (2019-2022), Durham University, University of Milano-Bicocca.
- Dr. Alejandro Benitez Llambay (2021-2022), University of Milano-Bicocca.
- Dr. Matteo Fossati (2018-2021), Durham University, University of Milano-Bicocca.
- Dr. Elisabeta Lusso (2017-2019), Junior Research Fellow, Durham University.
- Dr. Richard Bielby (2017-2019), Durham University.

Membership and Activities

2026- Co-director of DiSTEMLaB, Physics Department, University of Milano-Bicocca (www.distemlab.it)

2026- Spokesperson for Public Engagement, Physics Department, University of Milano-Bicocca

2026- Associate Member (elected), Sigma Xi, The Scientific Research Honour Society

2025- Tiger-team co-lead and member, wide-field spectroscopic telescope (WST) consortium

2023-2026 Co-lead of the outreach project "Un nuovo sguardo sul cielo di Milano" funded by NextGenerationEU

2023- Steering Committee, Bicocca Centre for Quantitative Cosmology, Dipartimenti di Eccellenza 2023

2022- Member of the International Astronomical Union

2021- Member of the Euclid Consortium

2021- Member of the MOSAIC/ELT Science working groups "First Light"and "Inventory of matter"

2021- Member of the Science Working Group, WEAVE survey

2020- Coordinator of Absorption Line Studies in the Quasar Working Group, WEAVE survey

2020,2025 Chair of PhD Admission Committee, Physics Department, University of Milano-Bicocca

2020 Panel Member, USA National Science Foundation

2018- Peer reviewer, Nature

2018- Peer reviewer, European Research Council

2017- Peer reviewer, Nature Astronomy

2016-2018 Member of Van Mildert College Council, Durham University

2016- HIRES/ELT Galaxy and IGM Working Group

2012- Peer reviewer, Astrophysical Journal

2012- Peer reviewer, Monthly Notices of the Royal Astronomical Society

2012- Peer reviewer, Astronomy and Astrophysics

2011-2012 Graduate Student Mentor, UCSC Astronomy & Astrophysics Department

2011-2015 Member, European Physical Society

2011-2012 Member, American Astronomical Society

2008-2015 Member, Società Italiana di Fisica

Refereed publications

1. Scialpi, M. et al. 2026, A&A in press (arXiv:2610.10687). *Cosmic Duets II. Growth and accretion in Gaia Multi-Peak dual AGN at Cosmic Noon.*
2. Travascio, A. et al. 2026, arXiv:2610.01454. *Constraining the density distribution of the cold Circumgalactic Medium of quasars at $z > 2$ through Lyα, HeII and Hα emission.*
2. Mangola, E.S., Pistis, F., **Fumagalli, M.**, Fossati, M. et al. 2026, arXiv:2609.28072. *Detection of metal absorption lines in quasar spectra: a neural network approach using U-Net.*
3. Oyarzún, G. et al. 2026, arXiv:2609.25221. *The KAGG survey: KCWI Analysis of Gas around Galaxies at $z \approx 3-5$.*
4. Tornotti, D., **Fumagalli, M.**, Fossati, M. et al. 2026, arXiv:2609.20938. *Cosmic web Lyα emission in a sample of overdense regions.*
5. Infurna, R.P et al.. 2026, arXiv:2609.15535. *ALMA-JELLY II. Constraining the radial profiles of the quenching timescales in ram-pressure-stripped galaxies.*
6. Herzog, G., Dutta, R., **Fumagalli, M.**. 2026, arXiv:2609.08654. *The environmental dependence of the circumgalactic medium in a high-resolution cosmological simulation.*

7. Ma, K. et al. 2026, MNRAS submitted, arXiv:2608.13153. *Forecast for the detectability of patchy hydrogen reionization in WEAVE-QSO measurements of the Lymanα forest power spectrum at redshift $z > 4$.*
8. Gentile, F. et al. 2026, A&A submitted, arXiv:2606.02687. *Euclid: Disky titans – surprisingly high star formation efficiency in two brightest group galaxies at $z \sim 0.75$.*
9. Negi, V. et al. 2026, ApJ submitted (arXiv:2606.12541). *DESI as sparse Integral Field Spectrograph I: Spatially resolved chemical enrichment in star-forming galaxies at $z < 0.1$*
10. Euclid Collaboration 2026, A&A in press, arXiv:2605.30440. *Euclid preparation. Probing galaxy evolution within cosmic voids in Euclid-like simulations.*
11. Wisz, M., et al. 2026, ApJ in press, arXiv:2604.16751. *The Qz5 Survey (II): Metallicity Evolution of Damped Lyα Systems Out to $z \sim 5$.*
12. Euclid Collaboration. 2026, arXiv:2601.10709. *Euclid preparation. 3D reconstruction of the cosmic web with simulated Euclid Deep spectroscopic samples.*
13. Galbiati, M., Tornotti, D., **Fumagalli, M.**, et al. 2025, arXiv:2512.16422. *MUSE Analysis of Gas around Galaxies (MAGG) - VII. Emission line galaxies near strong blended Lyα absorption systems at $z \gtrsim 3$.*
14. Chaudhary, R., Joshi, R., Das, S., **Fumagalli, M.** et al. 2025, A&A submitted (arXiv:2509.02672). *Baryonic Ecosystem IN Galaxies (BEINGMgII) – III. Cool gas reservoirs at $0.3 \leq z \leq 1.6$ in the Dark Energy Survey.*
15. Santo, E., **Fumagalli, M.**, Chang, S-K, et al. 2026, A&A, 707, A359. *The MUSE Ultra Deep Field (MUDF) VIII. The cool gas distribution surrounding galaxies at redshifts $z \sim 0.5 - 2$.*
16. Scialpi, M. et al. 2026, A&A 709, A236. *Cosmic Duets I. High-spatial resolution spectroscopy of dual and lensed AGN with MUSE.*
17. Tornotti, D., Fossati, M., **Fumagalli, M.**, et al. 2025, A&A, 704, 201. *Bayesian luminosity function estimation in multidepth datasets with selection effects: a case study for $3 < z < 5$ Lyα emitters.*
18. **Fumagalli, M.** 2024, arXiv:2409.00174. *The multiphase circumgalactic medium and its relation to galaxies: an observational perspective.*
19. Euclid Collaboration. 2025, A&A, 697, 1. *Euclid. I. Overview of the Euclid mission.*
20. Das, S., et al. 2025, A&A, 695, 207. *Baryonic Ecosystem in Galaxies (BEINGMgII) – II. Unveiling the Nature of Galaxies Harboring Cool Gas Reservoirs.*
21. Oyarzun, G.-A., et al. 2025, ApJ, 983, 10. *The Qz5 Survey (I): How the HI Mass Density of the Universe Evolves With Cosmic Time.*
22. Joshi, R., Das, S., **Fumagalli, M.** et al. 2025, A&A, 695, 206. *Baryonic Ecosystem in Galaxies (BEINGMgII). Host Galaxies of Ultra-strong MgII Absorbers in Subaru Hyper Suprime-Cam Survey.*
23. Travascio, A., et al. 2025, A&A, 694, 165. *X-ray view of a massive node of the Cosmic Web at $z \sim 3$ I. An exceptional overdensity of rapidly accreting SMBHs.*
24. Pensabene, A., Galbiati, M., **Fumagalli, M.** et al. 2025, A&A, 696, 33. *The MUSE Ultra Deep Field (MUDF) VII. Probing high-redshift gas structures in the surroundings of ALMA-identified massive dusty galaxies.*
25. Galbiati, M., et al. 2025, A&A, 696, 95. *Connecting the growth of galaxies to the large-scale environment in a massive node of the Cosmic Web at $z \sim 3$.*

26. Pistis, F., **Fumagalli, M.**, Fossati, M., et al. 2025, A&A, 698, 292. *Automated quasar continuum estimation using neural networks: a comparative study of deep-learning architectures.*
27. Wang, W., et al. 2025, Nature Astronomy, 9, 710. *A Giant Disk Galaxy Two Billion Years After The Big Bang.*
28. Mizener, A., et al. 2025, ApJ, 989, 114. *CLUES III: Do User Choices Impact The Results of SED Fitting? Tests of 'Off-The-Shelf' Stellar Population and Dust Extinction Models in the CLUES Sample.*
29. Chang, S.-J., et al. 2025, MNRAS, 542, 525. *Modeling Mg II resonance doublet spectra from galaxy haloes at $z \sim 1$.*
30. Tornotti, D., **Fumagalli, M.**, Fossati, M., et al. 2025, ApJL, 980, 43. *The MUSE Ultra Deep Field: A 5 Mpc stretch of the $z \approx 4$ cosmic web revealed in emission.*
31. Tornotti, D., **Fumagalli, M.**, Fossati, M., et al. 2025, Nat. Ast., 9, 577. *High-definition imaging of an extended filament connecting active quasars at cosmic noon.*
32. Galbiati, M., Dutta, R., **Fumagalli, M.** et al. 2024, A&A, 690, 7. *MUSE Analysis of Gas around Galaxies (MAGG) – VI. The cool and enriched gas environment of $z \geq 3$ Lyα emitters.*
33. Benitez-Llambay, A., Dutta, R., **Fumagalli, M.**, Navarro, J.F.. 2024, ApJ, 973, 61. *Examining the Nature of the Starless Dark Matter Halo Candidate Cloud-9 with Very Large Array Observations.*
34. D'Odorico, V., et al. 2024, Experimental Astronomy, 58, 21. *Galaxy Formation and Symbiotic Evolution with the Inter-Galactic Medium in the Age of ELT-ANDES.*
35. Dharmender, Joshi, R., **Fumagalli, M.** et al. 2024, A&A Letters, 692, 7. *Star-Formation in Neutral Hydrogen Gas Reservoirs at Cosmic Noon.*
36. Welsh, L., Cooke, R., **Fumagalli, M.**, et al. 2024, 691, 285. *A survey of extremely metal-poor gas at cosmic noon: evidence of elevated [O/Fe].*
37. Arnaudova, M.I. et al. 2024, A&A, 535, 2269. *WEAVE First Light Observations: Origin and Dynamics of the Shock Front in Stephan's Quintet.*
38. Dutta, R., **Fumagalli, M.**, Fossati, M., et al. 2024, 691, 236. *Metal line emission around $z < 1$ galaxies.*
39. Pruto, G., **Fumagalli, M.**, Rafelski, M., et al. 2024, 691, 222. *The stellar population of a $z \approx 3.25$ Lyα emitting group associated with a damped Lyα absorber.*
40. Revalski, M., et al. 2024, ApJ, 966, 228. *The MUSE Ultra Deep Field (MUDF). V. Characterizing the Mass-Metallicity Relation for Low Mass Galaxies at $z \approx 1 - 2$.*
41. Beckett, A. et al. 2024, ApJ, 974, 256. *The MUSE Ultra Deep Field (MUDF). VI. The relationship between galaxy properties and metals in the circumgalactic medium.*
42. Pensabene, A, et al. 2024, A&A, 684, 119. *ALMA survey of a massive node of the Cosmic Web at $z \sim 3$. I. Discovery of a large overdensity of CO emitters.*
43. Siressi, M., et al. 2024, AJ, 167, 166. *CLusters in the Uv as EngineS (CLUES). II. Sub-kpc scale outflows driven by stellar feedback.*
44. Dutta, R., Acebron, A., **Fumagalli, M.** et al. 2024, MNRAS, 528, 1895. *Probing coherence in metal absorption towards multiple images of strong gravitationally lensed quasars.*
45. Finn, M.K., et al. 2024, ApJ, 964, 13. *ALMA-LEGUS II: The Influence of Sub-Galactic Environment on Molecular Cloud Properties.*

46. Finn, M.K., et al. 2024, ApJ, 964, 12. *ALMA-LEGUS I: The Influence of Galaxy Morphology on Molecular Cloud Properties.*
47. Bortolini, G., et al. 2024, MNRAS, 527, 5339. *The spatially resolved star formation history of the dwarf spiral galaxy NGC 5474.*
48. Jin, S., et al. 2024, MNRAS, 530, 2688 *The wide-field, multiplexed, spectroscopic facility WEAVE: Survey design, overview, and simulated implementation.*
49. Stephenson, H.M.O, et al. 2023, MNRAS, 527, 7891. *Quasar Sightline and Galaxy Evolution (QSAGE) – III. The mass-metallicity and fundamental metallicity relation in $z \sim 2.2$ galaxies.*
50. de Beer, S. et al. 2023, MNRAS, 526, 1850 *Resolving the physics of Quasar Lyα Nebulae (RePhyNe): I. Constraining Quasar host halo masses through Circumgalactic Medium kinematics.*
51. Lusso, E., Nardini, E., **Fumagalli, M.** et al. 2023, MNRAS, 525, 4388. *The MUSE Ultra Deep Field (MUDF). IV. A pair of X-ray weak quasars at the heart of two extended Lyα nebulae.*
52. Jung, D.E. et al. 2023, ApJ, 954, 136. *Universal Upper End of the Stellar Initial Mass Function in the Young and Compact LEGUS clusters.*
53. Longobardi, A., Fossati, M., **Fumagalli, M.** et al. 2023, RASTI, 2, 470. *Towards an automatic approach to modelling the circumgalactic medium: new tools for mock making and fitting of metal profiles in large surveys.*
54. Welsh, L., Cooke, R., **Fumagalli, M.**, Pettini, M. 2023, MNRAS, 525, 527. *Towards ultra metal-poor DLAs: linking the chemistry of the most metal-poor DLA to the first stars.*
55. Teh, J.W. et al. 2023, MNRAS, 524, 1191. *Constraining the LyC escape fraction from LEGUS star clusters with SIGNALS HII region observations: A pilot study of NGC 628.*

56. Arrigoni Battaia, F. et al. 2023, A%A, 676, 51. *JCMT/SCUBA-2 uncovers an excess of 850μ m counts on megaparsec scales around high-redshift quasars. Characterization of the overdensities and their alignment with the quasars' Lyα nebulae.*
57. Urbano Stawinski, S.M et al. 2023, ApJ, 951, 135. *On the Metallicities and Kinematics of the Circumgalactic Media of Damped Lyα Systems at $z \sim 2.5$.*
58. Saccardi, A., Salvadori, S., D'Odorico, V. et al. 2023, ApJ, 948, 35, *Evidence of First Stars-enriched Gas in High-redshift Absorbers.*
59. Beckett, A., Morris, S.L., **Fumagalli, M.** et al. 2023, MNRAS, 521, 1113. *Modelling gas around galaxy pairs and groups using the Q0107 quasar triplet.*
60. Dutta, R., Fossati, M., **Fumagalli, M.** et al. 2023, MNRAS, 508, 4573. *Metal line emission from galaxy haloes at $z \approx 1$.*
61. Revalski, M., Rafelski, M., **Fumagalli, M.**, Fossati, M. et al. 2023, ApJS, 265, 40. *The MUSE Ultra Deep Field (MUDF) – III: Hubble Space Telescope WFC3 Grism Spectroscopy and Imaging.*
62. Galbiati, M., **Fumagalli, M.**, Fossati, M. et al. 2023, MNRAS, 524, 3474. *MUSE Analysis of Gas around Galaxies (MAGG) – V: Linking ionized gas traced by CIV and SiIV absorbers to Lyα emitting galaxies at $z \approx 3.0 - 4.5$.*
63. Cook, D.O., et al. 2023, MNRAS, 519, 3749. *Fraction of Stars in Clusters for the LEGUS Dwarf Galaxies.*

64. Luo, R., et al. 2023, MNRAS, 521, 6266. *Tracing the kinematics of the whole ram pressure stripped tails in ESO 137-001.*
65. Lofthouse, E., **Fumagalli, M.**, Fossati, M. et al. 2023, MNRAS, 518, 305. *MUSE Analysis of Gas around Galaxies (MAGG) – IV: The gaseous environment of $z \approx 3-4$ Lyα emitting galaxies.*
66. Herzog, G., Benitez-Llambay, A. **Fumagalli, M.** 2023, MNRAS, 518, 6305. *The present-day gas content of simulated field dwarf galaxies.*
67. Boselli, A. et al. 2023, A&A 669, 73. *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE).XIV. The main sequence relation in a rich environment down to $M_{star} \approx 10^6$ $M_\odot$.*
68. Siressi, M. et al. 2022, AJ, 164, 208. *CLusters in the UV as EngineS (CLUES). I. Survey Presentation and FUV Spectral Analysis of the Stellar Light.*
69. Beckett, A., Morris, S.L., **Fumagalli, M.** et al. 2022, MNRAS 517, 1020. *Signatures of extended discs and outflows in the circumgalactic medium using the Q0107 quasar triplet.*
70. Mintz, A., Rafelski, M., Jorgenson, R.A., **Fumagalli, M.** 2022, AJ, 164, 51. *Constraining the Size of the Circumgalactic Medium Using the Transverse Autocorrelation Function of C IV Absorbers in Paired Quasar Spectra.*
71. Robert, P.F., Murphy, M.T., O'Meara, J.M, Crighton, N.H.M, **Fumagalli, M.** 2022, MNRAS, 514, 3559. *Discovery of three new near-pristine absorption clouds at $z = 2.6-4.4$.*
72. Dalton, T., Morris, S.L., **Fumagalli, M.**, Gatuzz, E. 2022, MNRAS, 513, 822. *Probing the physical properties of the intergalactic medium using quasars.*
73. Welsh, L., Cooke, R.,**Fumagalli, M.**, Pettini, M.. 2022, ApJ, 929, 158. *Oxygen-enhanced extremely metal-poor DLAs: A signpost of the first stars?*
74. Lehner, N. et al. 2022, ApJ, 936, 156. *KODIAQ-Z: Metals and Baryons in the Cool Intergalactic and Circumgalactic Gas at 2.2<z<3.6.*
75. Arrigoni Battaia, F. et al. 2022, ApJ, 930, 72. *A Multiwavelength Study of ELAN Environments (AMUSE^2): Mass budget, satellites spin alignment and gas infall in a massive $z \sim 3$ quasar host halo.*
76. Pedrini, A. et al. 2022, MNRAS, 511, 5180. *MUSE sneaks a peek at extreme ram-pressure stripping events – V. Towards a complete view of the galaxy cluster A1367*
77. Nowotka, M. et al. 2022, A&A, 658, 77. *A Multiwavelength Study of ELAN Environments (AMUSE^2): Ubiquitous dusty star-forming galaxies associated with enormous Lyα nebulae on megaparsec scales.*
78. Orozco-Duarte, R. et al. 2022, MNRAS, 509, 522. *Synthetic photometry of OB star clusters with stochastically sampled IMFs: analysis of models and HST observations.*
79. Benitez-Llambay, A., **Fumagalli, M.** 2021, ApJL, 921, 9. *The Tail of Late-Forming Dwarf Galaxies in ΛCDM.*
80. Dutta, R., **Fumagalli, M.**, Fossati, M. et al. 2021, MNRAS, 508, 4573. *Metal-enriched halo gas across galaxy overdensities over the last 10 billion years.*
81. Dalton, T., Morris, S.L., **Fumagalli, M.**, Gatuzz, E. 2021, MNRAS, 508, 1701. *Probing the physical properties of the intergalactic medium using blazars.*
82. Menon, S.H. et al. 2021, MNRAS, 507, 5542. *The Dependence of the Hierarchical Distribution of Star Clusters on Galactic Environment.*

83. Beckett, A., Morris, S.L., **Fumagalli, M.** et al. 2021, MNRAS, 506, 2574. *The relationship between gas and galaxies at $z < 1$ using the Q0107 quasar triplet.*
84. Della Bruna, L. et al., 2021, A&A, 663, 2. *Studying the ISM at ~ 10 pc scale in NGC 7793 with MUSE – II. Constraints on the oxygen abundance and ionising radiation escape.*
85. Fossati, M., **Fumagalli, M.**, Lofthouse, E.K. et al. 2021, MNRAS, 503, 3044. *MUSE Analysis of Gas around Galaxies (MAGG) - III: The gas and galaxy environment of $z = 3 - 4.5$ quasars.*
86. Dalton, T., Morris, S.L., **Fumagalli, M.**. 2021, MNRAS, 502, 5981. *Probing the physical properties of the intergalactic medium using gamma-ray bursts.*
87. Berg, T.A.M, **Fumagalli, M.**, D'Odorico, V. et al. 2021, MNRAS, 502, 4009. *Sub-damped Lyman alpha systems in the XQ-100 survey II – Chemical evolution at $2.4 < z < 4.3$.*
88. Joshi, R., **Fumagalli, M.**, Srianand, R. et al. 2021, ApJ, 908, 129. *Discovery of a damped Lyα galaxy at $z \sim 3$ towards the quasar SDSS J011852+040644.*
89. Welsh, L., Cooke, R.,**Fumagalli, M.**. 2021, MNRAS, 500, 5214. *The stochastic enrichment of Population II stars.*
90. Della Bruna, L., Adamo, A., Bik A., **Fumagalli M.** et al. 2021, A&A, 652, 6. *Studying the ISM at 10 pc scale in NGC 7793 with MUSE – I. Data description and properties of the ionised gas.*
91. Dutta, R., **Fumagalli, M.**, Fossati, M. et al. 2020, MNRAS, 499, 5022. *MUSE Analysis of Gas around Galaxies (MAGG) - II: Metal-enriched halo gas around z 1 galaxies.*
92. Decataldo, D., Lupi, A., Ferrara, A., Pallottini, A. **Fumagalli, M.** 2020, MNRAS, 497, 4718. *Shaping the structure of a GMC with radiation and winds.*
93. **Fumagalli, M.**, Fotopoulou, S., Thomson, L. 2020, MNRAS, 498, 1951. *Detecting neutral hydrogen at $z > 3$ in large spectroscopic surveys of quasars.*
94. Stott, J.P. et al. 2020, MNRAS, 497, 3083. *Quasar Sightline and Galaxy Evolution (QSAGE) survey – II. Galaxy overdensities around UV luminous quasars at $z = 1 - 2$.*
95. Bielby, R., **Fumagalli, M.**, Fossati, M. et al. 2020, MNRAS, 493, 5336. *Into the Lyα jungle: exploring the circumgalactic medium of galaxies at $z \sim 4 - 5$ with MUSE.*
96. Cooke, R., Welsh, L., **Fumagalli, M.**, Pettini, M. 2020, MNRAS, 494, 4884. *A limit on Planck-scale froth with ESPRESSO.*
97. Welsh, L., Cooke, R., **Fumagalli, M.**, Pettini, M. 2020, MNRAS, 494, 1411. *A bound on the 12C/13C ratio in near-pristine gas with ESPRESSO.*
98. Buie, E., **Fumagalli, M.**, Scannapieco, E. 2020, 890, 33. *Interpreting Observations of Absorption Lines in the Circumgalactic Medium with a Turbulent Medium.*
99. Lofthouse, E.K., **Fumagalli, M.**, Fossati, M. et al. 2020, MNRAS, 491, 2057. *MUSE Analysis of Gas around Galaxies (MAGG) – I: Survey design and the environment of a near pristine gas cloud at z 3.5.*
100. Sykes, C., **Fumagalli, M.**, Cooke, R., Theuns, T. 2020, MNRAS, 492, 2151. *Determining the primordial helium abundance and UV background using fluorescent emission in star-free dark matter haloes.*
101. Fossati, M., **Fumagalli, M.**, Lofthouse, E.K. et al. 2019, MNRAS, 490, 1451. *The MUSE Ultra Deep Field (MUDF). II. Survey design and the gaseous properties of galaxy groups at $0.5 < z < 1.5$.*

102. Umehata, H., **Fumagalli, M.**, Smail, I. et al. 2019, Science, 366, 97. *Gas filaments of the cosmic web located around active galaxies in a proto-cluster.*

103. Becker, G.D. et al. 2019, ApJ, 883, 163. *The Evolution of OI over $3.2 < z < 6.5$: Reionization of the Circumgalactic Medium.*

104. Jachym, P. et al. 2019, ApJ, 883, 145. *ALMA unveils widespread molecular gas clumps in the ram pressure stripped tail of the Norma jellyfish galaxy.*

105. **Fumagalli, M.** 2019, Nature Astronomy, 3, 796. *Thirsty galaxies thriving on gas streams.*

106. Sykes, C., **Fumagalli, M.**, Cooke, R., Theuns, T., Benitez-Llambay, A. 2019, MNRAS, 487, 609. *Fluorescent rings in star-free dark matter haloes.*

107. Mackenzie, R., **Fumagalli, M.**, Theuns, T. et al. 2019, MNRAS, 487, 5070. *Linking gas and galaxies at high redshift: MUSE surveys the environments of six damped Lyα galaxies at $z \sim 3$.*

108. Welsh, L., Cooke, R., **Fumagalli, M.** 2019, MNRAS, 487, 3363. *Modelling the chemical enrichment of Population III supernovae: The origin of the metals in near-pristine gas clouds.*

109. Bielby, R.M. et al. 2019, MNRAS, 86, 21. *Quasar Sightline and Galaxy Evolution (QSAGE) Survey - I. The Galaxy Environment of OVI Absorbers up to $z = 1.4$ around PKS 0232-04.*

110. Lusso, E., **Fumagalli, M.**, Fossati, M., et al. 2019, MNRAS, 485, 62. *The MUSE Ultra Deep Field (MUDF). I. Discovery of a group of Lyα nebulae associated with a bright $z \approx 3.23$ quasar pair.*

111. Furniss, A., Worseck, G., **Fumagalli, M.** et al. 2019, AJ, 157, 41. *Spectroscopic Redshift of the Gamma-Ray Blazar B2 1215+30 from Lyα Emission.*

112. Cook, D.O. et al. 2019, MNRAS, 484, 4897. *Star Cluster Catalogs for the LEGUS Dwarf Galaxies.*

113. Fossati, M., **Fumagalli, M.**, Gavazzi, G. et al. 2019, MNRAS, 484, 2212. *MUSE sneaks a peek at extreme ram-pressure stripping events - IV. Hydrodynamic and gravitational interactions in the Blue Infalling Group.*

114. Grasha, K. et al. 2019, 483, 4707. *The Spatial Relation between Young Star Clusters and Molecular Clouds in M 51 with LEGUS.*

115. P. Frédéric Robert et al. 2019, MNRAS, 483, 2736. *Exploring the origins of a new, apparently metal-free gas cloud at $z = 4.4$.*

116. Jauzac, M. et al. 2019, MNRAS, 483, 3082. *The core of the massive cluster merger MACS J0417.5-1154 as seen by VLT/MUSE.*

117. Arrigoni Battaia, F., Chen, C.-C., **Fumagalli, M.** et al. 2018, A&A, 620, 202. *Over-density of submillimeter galaxies around the z=2.3 MAMMOTH-1 nebula.*

118. Boselli, A. et al. 2018, A&A, 620, 164. *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE).IV. A tail of Ionised Gas in the Merger Remnant NGC 4424.*

119. Krumholz, M. R., Adamo, A., **Fumagalli, M.**, Calzetti, D. 2019, MNRAS, 482, 3550. *SLUG IV: A Novel Forward-Modelling Method to Derive the Demographics of Star Clusters.*

120. Caruso, D., Haardt, F., **Fumagalli, M.**, Cantalupo, S. 2019, MNRAS, 482, 2833. *MCMC determination of the cosmic UV background at $z \approx 0$ from Hα fluorescence.*

121. Cooke, R. & **Fumagalli, M.** 2018, Nature Astronomy, 2, 957. *Measurement of the primordial helium abundance from the intergalactic medium.*

122. Krogager, J.-K. et al. 2018, A&A, 619, 142. *Dissecting cold gas in a high-redshift galaxy using a lensed background quasar.*

123. Grasha, K. et al. 2018, MNRAS, 481, 1016. *Connecting Young Star Clusters to CO Molecular Gas in NGC 7793 with ALMA-LEGUS.*

124. Ashworth, G., **Fumagalli, M.**, Adamo, A., Krumholz, M.R. 2018, MNRAS, 480, 3091A. *Theoretical predictions for IMF diagnostics in UV spectroscopy of star clusters.*

125. Hunter, D. et al. 2018, AJ, 156, 21. *A comparison of young star properties with local galactic environment for LEGUS/LITTLE THINGS dwarf irregular galaxies.*

126. Boselli, A. et al. 2018, A&A, 615, 114. *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE).III. Star formation in the stripped gas of NGC 4254.*

127. Chehade, B. et al. 2018, MNRAS, 478, 1649. *Two more, bright, $z > 6$ quasars from VST ATLAS and WISE .*

128. Boselli, A. et al. 2018, A&A, 614, 56. *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE).I. Introduction to the Survey.*

129. Fossati, M. et al. 2018, A&A, 614, 57. *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE). III. Constraining the quenching time in the stripped galaxy NGC 4330.*

130. Lusso, E., **Fumagalli, M.**, Rafelski, M. et al. 2018, ApJ, 860, 41. *The spectral and environment properties of $z \sim 2.0 - 2.5$ quasar pairs.*

131. Findlay, J.R. et al. 2018, ApJS, 236, 44. *Quasars probing quasars X: The quasar pair spectral database.*

132. Messa, M. et al. 2018, MNRAS, 477, 1683. *The Young Star Cluster population of M51 with LEGUS: II. Testing environmental dependencies.*

133. Kahre, L. et al. 2018, ApJ, 855, 133. *Extinction Maps and Dust-to-Gas Ratios in Nearby Galaxies.*

134. Gavazzi, G., Consolandi, G., Pedraglio, S., Fossati, M., **Fumagalli, M.**, Boselli, A. 2018, A&A, 611, 28. *Hα imaging observations of early-type galaxies from the ATLAS3D survey.*

135. Hunter, D. et al. 2018, ApJ, 855, 7. *A study of two dwarf irregular galaxies with asymmetrical star formation distributions.*

136. Sabbi, E. et al. 2018, ApJS, 235, 23. *The resolved stellar populations in the LEGUS galaxies.*

137. Messa, M. et al. 2018, MNRAS, 473, 996. *The Young Star Cluster population of M51 with LEGUS: I. A comprehensive study of cluster formation and evolution.*

138. Consolandi, G., Gavazzi, G., Fossati, M., **Fumagalli, M.**, Boselli, A., Yagi, M., Yoshida, M. et al. 2017, A&A, 606, 83. *MUSE sneaks a peek at extreme ram-pressure events - III. Tomography of UGC 6697, a massive galaxy falling into Abell 1367.*

139. **Fumagalli, M.**, Mackenzie, R., Trayford, J. et al. 2017, MNRAS, 471, 3686. *Witnessing galaxy assembly in an extended $z \approx 3$ structure.*

140. Grasha, K. et al. 2017, ApJ, 842, 25. *Hierarchical Star Formation in Turbulent Media: Evidence from Young Star Clusters.*

141. Ashworth, G., **Fumagalli, M.**, Krumholz, M.R. et al. 2017, MNRAS, 469, 2464. *Exploring the IMF of star clusters: a joint SLUG and LEGUS effort.*

142. Ryon, J.E. et al. 2017, ApJ, 841, 92. *Effective Radii of Young, Massive Star Clusters in Two LEGUS Galaxies.*

143. Adamo, A. et al. 2017, ApJ, 841, 131. *Legacy ExtraGalactic UV Survey with The Hubble Space Telescope. Stellar cluster catalogues and first insights into cluster formation and evolution in NGC 628.*

144. Grasha, K. et al. 2017, ApJ, 840, 113. *The Hierarchical Distribution of the Young Stellar Clusters in Six Local Star Forming Galaxies.*

145. Bielby, R., Crighton, N.H.M, **Fumagalli, M.** et al. 2017, MNRAS, 468, 1373. *Probing the intra-group medium of a $z = 0.28$ galaxy group.*

146. Swinbank, M. et al. 2017, MNRAS, 467, 3140. *Angular momentum evolution of galaxies over the past 10 Gyr: A MUSE and KMOS dynamical survey of 400 star-forming galaxies from $z = 0.3 - 1.7$.*

147. **Fumagalli, M.**, Haardt, F., Theuns, T., Morris, S.L., Cantalupo, S., Madau, P., Fossati, M. 2017, MNRAS, 467, 4802. *A measurement of the $z = 0$ UV background from Hα fluorescence.*

148. Prochaska et al. 2017, ApJ, 837, 169. *The COS-Halos Survey: Metallicities in the Low-Redshift Circumgalactic Medium.*

149. Lehner, N., O'Meara, J.M., Howk, J.C., Prochaska, J.X., **Fumagalli, M.** 2016, ApJ, 833, 283. *The Cosmic Evolution of the Metallicity Distribution of Ionized Gas Traced by Lyman Limit Systems.*

150. Toy, V.L. et al. 2016, ApJ, 832, 175. *Exploring Damped Lyman-α System Host Galaxies using Gamma-ray Bursts.*

151. **Fumagalli, M.**, Cantalupo, S., Dekel, A., Morris, S.L., O'Meara, J.M, Prochaska, J.X., Theuns, T. 2016, MNRAS, 462, 1978. *MUSE searches for galaxies near very metal-poor gas clouds at $z \sim 3$: new constraints for cold accretion models.*

152. Rafelski, M., Gardner, J.P., **Fumagalli, M.** et al. 2016, ApJ, 825, 87. *The Star Formation Rate Efficiency of Neutral Atomic-dominated Hydrogen Gas in the Outskirts of Star Forming Galaxies from $z \sim 1$ to $z \sim 3$.*

153. Consolandi, G., Gavazzi, G., **Fumagalli, M.** et al. 2016, A&A, 591, 38. *Robust automatic photometry of local galaxies from SDSS. Dissecting the color magnitude relation with color profiles.*

154. Finn, C. et al. 2016, MNRAS, 460, 590. *On the connection between the metal-enriched intergalactic medium and galaxies: an OVI-galaxy cross-correlation study at $z < 1$.*

155. Archambault, S. et al. 2016, AJ, 151, 142. *Upper limits from five years of blazar observations with the VERITAS Cherenkov telescopes.*

156. Boselli, A. et al. 2016, A&A, 587, 68. *Spectacular tails of ionised gas in the Virgo cluster galaxy NGC 4569*

157. Grasha, K. et al. 2015, ApJ, 815, 93. *The Spatial Distribution of the Young Stellar Clusters in the Star Forming Galaxy NGC 628.*

158. **Fumagalli, M.**, O'Meara, J.M., Prochaska, J.X. 2016, MNRAS, 455, 4100. *The physical properties of $z > 2$ Lyman limit systems: new constraints for feedback and accretion models.*

159. Fossati, M., **Fumagalli, M.**, Boselli, A. et al. 2016, MNRAS, 455, 2028. *MUSE sneaks a peek at extreme ram-pressure stripping events. II. The physical properties of the gas tail of ESO137-001.*

160. Farina, E., **Fumagalli, M.**, Decarli, R. et al. 2016, MNRAS, 455, 618. *The Cluster-Scale Environment of PKS 2155-304*.

161. Krumholz, M. R., Adamo, A., **Fumagalli, M.**, et al. 2015, ApJ, 812, 147. *Star Cluster Properties in Two LEGUS Galaxies Computed with Stochastic Stellar Population Synthesis Models*.

162. Calzetti, D. et al. 2015, ApJ, 811, 75. *The Brightest Young Star Clusters in NGC 5253*.

163. Prochaska, J.X. et al. 2015, ApJS, 221, 2. *The Keck+Magellan Survey for Lyman Limit Absorption III: Sample Definition and Column Density Measurements*.

164. Crighton, N. et al. 2015, MNRAS, 452, 217. *The Neutral Hydrogen Cosmological Mass Density at* $z = 5$.

165. Krumholz, M., **Fumagalli, M.**, da Silva, R., Rendahl, T., Parra, J. 2015, MNRAS, 452, 1447.*SLUG – Stochastically Lighting Up Galaxies. III: A Suite of Tools for Simulated Photometry, Spectroscopy, and Bayesian Inference with Stochastic Stellar Populations*.

166. Gavazzi, G. et al. 2015, A&A, 580, 116. *Halpha3: an Halpha imaging survey of HI selected galaxies from ALFALFA . VI. The role of bars in quenching star formation from z = 3 to the present epoch*.

167. Carnall, A. C. et al. 2015, MNRAS, 451, 16.*Two bright* $z > 6$ *quasars from VST ATLAS and a new method of optical plus mid-infra-red colour selection*.

168. Cucchiara, A., **Fumagalli, M.**, Rafelski, M., Kocevski, D., Prochaska, J.X., Cooke, R.J., Becker, G.D. 2015, ApJ, 804, 51.*Unveiling the Secrets of Metallicity and Massive Star Formation Using DLAs along Gamma-ray Bursts*.

169. Gavazzi, G. et al. 2015, A&A, 576, 16. *Halpha3: an Halpha imaging survey of HI selected galaxies from ALFALFA . V. The Coma supercluster survey completion*.

170. Calzetti, D. et al. 2015, AJ, 149, 51. *Legacy ExtraGalactic UV Survey (LEGUS) with The Hubble Space Telescope. I. Survey Description*.

171. **Fumagalli, M.**, O'Meara, J.M., Prochaska, J.X., Rafelski, M., Kanekar, N. 2015, MNRAS, 446, 3178. *Directly imaging damped Ly*α *galaxies at* $z > 2$*. III: The star formation rates of neutral gas reservoirs at* $z \sim 2.7$.

172. Crighton, N. et al. 2015, MNRAS, 446, 18. *Metal-enriched, sub-kiloparsec gas clumps in the circumgalactic medium of a faint* $z = 2.5$ *galaxy*.

173. **Fumagalli, M.**, Fossati, M., Hau, G. et al. 2014, MNRAS, 445, 4335.*MUSE sneaks a peek at extreme ram-pressure stripping events. I. A kinematic study of the archetypal galaxy ESO137-001*.

174. Aliu, E. et al. 2014, ApJ, 797, 89. *Investigating Broadband Variability of the TeV Blazar 1ES 1959+650*.

175. Boselli, A. et al. 2014, A&A, 570, 69. *The GALEX Ultraviolet Virgo Cluster Survey (GUViCS). IV: The role of the cluster environment on galaxy evolution*

176. Worseck, G. et al. 2014, MNRAS, 445, 1745.*The Giant Gemini GMOS survey of* $z > 4.4$ *quasars. I: Measuring the mean free path across cosmic time.*

177. da Silva, R.L., **Fumagalli, M.**, Krumholz, M. 2014, MNRAS, 444, 3275. *SLUG - Stochastically Lighting Up Galaxies. II: Quantifying the Effects of Stochasticity on Star Formation Rate Indicators*.

178. **Fumagalli, M.**, O'Meara, J.M., Prochaska, J.X., Kanekar, N., Wolfe, A. 2014, MNRAS, 444, 1282. *Directly imaging damped Lyα galaxies at $z > 2$. II: Imaging and spectroscopic observations of 32 quasar fields*.

179. Lusso, E. et al. 2014, MNRAS, 441, 316. *The nature of massive black hole binary candidates: II. Spectral energy distribution atlas.*

180. Finn, C. et al. 2014, MNRAS, 440, 3317. *A compact, metal-rich, kpc-scale outflow in FBQS J0209-0438: Detailed diagnostics from HST/COS extreme UV observations.*

181. da Silva, R.L., Krumholz, M., **Fumagalli, M.**, Fall, M. 2014, MNRAS, 438, 2355. *An Analytic Method to Compute Star Cluster Luminosity Statistics.*

182. Wright, E. et al., 2014, AJ, 147, 61. *The First AllWISE Proper Motion Discovery: WISEA J070720.50+170532.7.*

183. Rafelski, M., Neeleman, M., **Fumagalli, M.**, Wolfe, A.M., Prochaska, J.X. 2014, ApJL, 782, 29. *The Rapid Decline in Metallicity of Damped Ly-α Systems at $z \sim 5$.*

184. Prochaska, J.X., Madau, P., O'Meara, J.M., **Fumagalli, M.** 2014, MNRAS, 438, 476. *Towards a Unified Description of the Intergalactic Medium at Redshift $z \sim 2.5$.*

185. **Fumagalli, M.**, Hennawi, J., Prochaska, J.X., Kasen, D., Dekel, A., Ceverino, D., Primack, J. 2014, ApJ, 780, 74. *Confronting Simulations of Optically Thick Gas in Massive Halos with Observations at $z = 2 - 3$.*

186. VERITAS collaboration et al., 2013, ApJ, 779, 92. *Long term observations of B2 1215+30 with VERITAS.*

187. **Fumagalli, M.**, O'Meara, J.M., Prochaska, J.X., Worseck, G. 2013, ApJ, 775, 78. *Dissecting the properties of optically-thick hydrogen at the peak of cosmic star formation history*.

188. Decarli, R., Dotti, M., **Fumagalli, M.**, et al. 2013, MNRAS, 433, 1492. *The nature of massive black hole binary candidates: I. Spectral properties and evolution*.

189. Furniss, A., **Fumagalli, M.**, Falcone, A., Williams, D. A. 2013, ApJ, 770, 109. *The Blazar Emission Environment: Insight from Soft X-ray Absorption*.

190. Furniss, A. et al. 2013, ApJ, 768, L31. *The Firm Redshift Lower Limit of the Most Distant TeV-Detected Blazar PKS 1424+240*.

191. Fossati, M. et al. 2013, A&A, 553, 91. *Halpha3: an Halpha imaging survey of HI selected galaxies from ALFALFA. IV. The structure of galaxies in the Local and Coma Superclusters*.

192. Gavazzi, G. et al. 2013, A&A, 553, 90. *Halpha3: an Halpha imaging survey of HI selected galaxies from ALFALFA. III. Nurture shapes up the Hubble sequence in the Great Wall*.

193. Gavazzi, G., **Fumagalli, M.**, Fossati, M. et al. 2013, A&A, 553, 89. *Halpha3: an Halpha imaging survey of HI selected galaxies from ALFALFA. II. The star formation properties of galaxies in the Virgo cluster and surroundings*.

194. Furniss, A., **Fumagalli, M.**, Danforth, C., Williams, D., & Prochaska, X. 2013, ApJ, 766, 35. *On the Redshift of the Very High Energy Blazar 3C66A*.

195. Farina, E., Montuori, C., Decarli, R., **Fumagalli, M.** 2013, MNRAS, 431, 1019. *Caught in the Act: Discovery of a Physical Quasar Triplet*.

196. Aliu, E. et. al 2012, ApJ, 759, 102. *VERITAS Observations of Six Bright, Hard-Spectrum Fermi-LAT Blazars*.

197. **Fumagalli, M.**, Furniss, A., O'Meara, J., Prochaska, X., Williams, D., Farina, E. 2012, A&A, 545, 68. *On the redshift of the blazar PKS0447-439*.

198. Fossati, M., Gavazzi, G., Boselli, A., **Fumagalli, M.** 2012, A&A, 544, A128. *65 kpc of ionized gas trailing behind NGC 4848 during its first crossing of the Coma cluster*.

199. Gavazzi, G., **Fumagalli, M.**, Galardo, V., et al. 2012, A&A, 545, 16. *Hα^3: Hα imaging survey of HI selected galaxies from ALFALFA. I. Catalogue in the local supercluster*.

200. **Fumagalli, M.**, Dessauges-Zavadsky, M., Furniss, A., et al. 2012, MNRAS, 424, 2276. *A search of CO emission lines in blazars: the low molecular gas content of BL Lac objects compared to quasars*.

201. Arrigoni Battaia, F. et al. 2012, A&A, 543 A112. *Stripped gas as fuel for newly formed HII regions in the encounter between VCC1249 and M49: a unified picture from NGVS and GUViCS*.

202. da Silva, R.L., **Fumagalli, M.**, & Krumholz, M. 2012, ApJ, 745, 145. *SLUG - Stochastically Lighting Up Galaxies I: Methods and Validating Tests*.

203. **Fumagalli, M.**, O'Meara, J.M., & Prochaska, J.X. 2011, Science, 334, 1245. *Detection of pristine gas two billion years after the Big Bang*.

204. Barth, A. et al. 2011, ApJ, 743, L4. *The Lick AGN monitoring project 2011: reverberation mapping of Markarian 50*.

205. **Fumagalli, M.**, da Silva, R.L., & Krumholz, M. 2011, ApJ, 741, L26. *Stochastic star formation and a (nearly) uniform stellar initial mass function*.

206. Aliu, E. et al. 2011, ApJ, 742, 127. *Multiwavelength Observations of the Previously Unidentified Blazar RX J0648.7+1516*.

207. Cucchiara, A. et al. 2011, ApJ, 743, 154. *Constraining GRB Emission Physics with Extensive Early-Time, Multiband Follow-up*.

208. **Fumagalli, M.**, Prochaska, J.X., Kasen, D., Dekel, A., Ceverino, D., & Primack, J.R. 2011, MNRAS, 418, 1796. *Absorption line systems in simulated galaxies fed by cold streams*.

209. Barth, A. et al. 2011, ApJ, 732, 121. *Broad-line Reverberation in the Kepler-field Seyfert Galaxy Zw 229-015*.

210. Abdo, A.A. et al. 2011, ApJ, 726, 43. *Multi-wavelength Observations of the Flaring Gamma-ray Blazar 3C 66A in 2008 October*.

211. **Fumagalli, M.**, O'Meara, J.M., Prochaska, J.X., & Kanekar, N. 2010, MNRAS, 408, 362. *Directly imaging damped Lyman-α galaxies at $z > 2$ - I. Methodology and first results*.

212. **Fumagalli, M.**, Krumholz, M.R., & Hunt, L.K. 2010, ApJ, 722, 919. *Testing models for molecular gas formation in galaxies: hydrostatic pressure or gas and dust shielding?*

213. **Fumagalli, M.**, Krumholz, M.R., Prochaska, J.X., Gavazzi, G., & Boselli, A. 2009, ApJ, 697, 1811 *Molecular hydrogen deficiency in HI-poor galaxies and its implications for star formation*.

214. **Fumagalli, M.**, & Gavazzi, G. 2008, A&A, 490, 571. *The relationship between gas content and star formation rate in spiral galaxies. Comparing the local field with the Virgo cluster*.

215. Gavazzi, G. et al. 2008, A&A, 482, 43. *HI content and other structural properties of galaxies in the Virgo cluster from the Arecibo Legacy Fast ALFA Survey*.

Non-refereed publications

1. Richard, J. et al. 2019, arXiv:1906.01657. *BlueMUSE: Project Overview and Science Cases*.
2. DESI collaboration 2016, arXiv:1611.00037. *The DESI Experiment Part II: Instrument Design*.
3. DESI collaboration 2016, arXiv:1611.00036. *The DESI Experiment Part I: Science,Targeting, and Survey Design*.
4. Pieri, M. et al. 2016, Proceedings of the SF2A conference, Lyon, 2016. *WEAVE-QSO: A Massive Intergalactic Medium Survey for the William Herschel Telescope*.
5. **Fumagalli, M.** 2014, MmSAI, 85, 355. *Metal abundances in the high-redshift intergalactic medium*.
6. **Fumagalli, M.** 2012, Ph.D. dissertation, University of California, Santa Cruz. *Food for stars: the role of hydrogen in the formation and evolution of galaxies*.
7. **Fumagalli, M.**, da Silva, R., Krumholz, M., & Bigiel, F. 2011, Astronomical Society of the Pacific Conference Series, 440, 155. *SLUG: A New Way to Stochastically Light Up Galaxies*.
8. **Fumagalli, M.** 2008, MSc thesis, Università Milano-Bicocca. *High resolution multifrequency analysis of gas behavior and star formation in spiral galaxies*.
9. **Fumagalli, M.** 2006, BSc thesis, Università Milano-Bicocca. *Impact of low frequencies measurements on the knowledge of spectral distortions expected for Cosmic Microwave Background Radiation*.